

SOKKANAEM: Change-Gated State-Space Modeling with Cached Readout for Streaming Video Depth

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Abstract

Streaming video depth requires dense predictions on every frame, even when much of the image changes little. We present SOKKANAEM, a 4.19-million-parameter recurrent model that combines patch-change gating, selective state-space updates, temporal and spatial output caches, and a dense depth decoder. Binary gating preserves inactive temporal states, while periodic refresh and an all-active fallback limit reuse duration and handle high activity. We evaluate the frozen model with same-weight controls that separate state carry, retained-state readout, spatial context and output reuse. On four reserved TUM sequences, the reference model obtains clip scale-shift AbsRel 0.1869 at 256 frames, compared with 0.1837 for dense carry and 0.1849 for MiDaS Small. Its motion-referenced TCE is 0.0352 versus 0.0640 for MiDaS, although its scale variation is not uniformly smaller. The measured development PNG-to-depth latency is 7.496 ms/frame versus 7.521 for dense carry. Thus, computational reuse does not yet establish substantial pipeline acceleration. Conditional error and cache-age analysis explains the design trade-offs. The contribution is a compact streaming architecture and a controlled characterization of its accuracy, temporal behavior and implementation cost, rather than a claim of universal performance superiority.

Keywords: state-space model; conditional computation; video depth; error propagation; cache refresh; reproducible evaluation

Table 1. Width fallback smoke test (not manuscript content).

	Value
Wide test cell	1
Received:	
Revised:	
Accepted:	
Published:	
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